

An Improved Upper Bound for Diagonal Ramsey Numbers

Sunghyeon Jo

Georgia Institute of Technology and QED Audit

sj65@gatech.edu

Abstract

We prove that $R(k, k) \leq 3.68395^k$ for all sufficiently large k . The proof combines a weighted extension of the candidate lemma of Gupta, Ndiaye, Norin and Wei with a summation over subsets of a maximum red clique. The complete proof has been formally verified in Lean 4.

1 Introduction

The Ramsey number $R(k, \ell)$ is the least integer N such that every red–blue colouring of the edges of K_N contains a red K_k or a blue K_ℓ . The classical bound of Erdős and Szekeres [5] is $R(k, \ell) \leq \binom{k+\ell-2}{k-1}$. In particular, $R(k, k) \leq 4^{k+o(k)}$. Successive improvements by Thomason [14], Conlon [3] and Sah [13] improved the subexponential factor, but left the exponential base unchanged.

Campos, Griffiths, Morris and Sahasrabudhe [2] proved that $R(k, k) \leq (4 - \varepsilon)^k$ for some absolute constant $\varepsilon > 0$. Their proof constructs large monochromatic books. A red book consists of a red clique S and a disjoint set T , with all edges between S and T red; blue books are defined similarly. Off-diagonal Ramsey bounds then complete a monochromatic clique. Gupta, Ndiaye, Norin and Wei [6] refined this argument, obtaining $R(k, k) \leq (4e^{-0.14/e})^{k+o(k)}$, where $4e^{-0.14/e} = 3.7992\dots$, together with an explicit off-diagonal bound. Further iterations of their optimization have been reported numerically: an additional iteration with base 3.78233, recorded as unverified in [6, Remark 17], and an eight-round iteration with base 3.776 in the thesis of Ndiaye [11].

Neisler, Shin and Sukhatankar [12] then made the bootstrap self-consistent: the bound being proved itself defines the admissible parameters of every round, so the whole iteration of these rules collapses to a single system of inequalities. Optimizing rate functions within that system, they proved $R(k, k) \leq 3.769^{k+o(k)}$, with a kernel-checked Lean proof of the decimal bound 3.7690^k .

Our proof starts from the same off-diagonal estimate of [6]. We introduce a weight θ into the candidate lemma and extend it by induction using density estimates established in earlier stages. We prove the following.

Theorem 1.1. *For all sufficiently large positive integers k ,*

$$R(k, k) \leq 3.68395^k.$$

The numerical inequalities and the arguments connecting them to Theorem 1.1 have been formalized in the Lean proof assistant [4]. The proof depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The complete development and its verification records are available in the accompanying repository [9]. It builds on the formalization [7] of the Ramsey definitions and of the input estimate of [6], in the copy distributed with [12]. Theorem 1.1 improves the strongest previously established upper bound for $R(k, k)$, the kernel-checked 3.7690^k of [12].

Numerical claims with bases 3.7296 and 3.6961 appear in version 1 of the HorizonMath benchmark [15, 8]. Appendix C shows that the released checks do not establish all hypotheses of the density theorem on which these claims rely. This concerns their justification, not the possibility of proving either bound.

1.1 Outline of the proof

We first explain how to strengthen the candidate lemma. Consider disjoint sets X and Y with many red edges between them. Following [6], we seek one of three outcomes: a red K_k in $X \cup Y$, a blue K_t in X , or a blue K_ℓ in Y . We prove a weighted version of this lemma, with a size condition on $|X|^\theta |Y|$ rather than $|X||Y|$. The parameter $\theta > 0$ allows the sizes of the two sets to contribute differently to this condition.

The next step is to extend this estimate to larger values of t . First discard the vertices of X with few red neighbours in Y . Every remaining vertex has a prescribed minimum red degree into Y , and this remains true when X is replaced by any of its subsets. We now keep Y fixed and induct on t . If X has sufficiently large red density, a previously proved density bound gives a red K_k or a blue K_t . Otherwise, some vertex has a large blue neighbourhood in X . The induction applies to that neighbourhood together with Y ; a blue K_{t-1} in the neighbourhood extends by the chosen vertex. Thus the initial estimate can use the size of Y without losing its contribution during the induction.

The density bound used in this argument may itself have been proved by an earlier such induction. Repeated application gives an improved off-diagonal bound. In exponential coordinates, the size lost at a blue-neighbourhood step becomes a lower bound on the derivative of an exponent function. The strict inequalities in Theorem 4.4 allow us to pass from these functions to integer clique sizes, including where successive pieces of a function meet.

It remains to turn the off-diagonal bound into a diagonal one. Choose a colour of density at least $1/2$, say red, and pass to a subgraph of almost that minimum red degree. Let S be a largest red clique. For a subset $A \subseteq S$, the common red neighbourhood of A cannot contain a red clique of order greater than $|S| - |A|$, since S is largest. When the off-diagonal bound has an affine exponent, these neighbourhood bounds can be summed over all subsets A at once, with a weight geometric in $|A|$; convexity bounds the same sum from below in terms of the order of the graph. The same set S controls both estimates; we need not know $|S|$ in advance.

All logarithms are natural. To state the resulting inequality, put $\Gamma = \limsup_{k \rightarrow \infty} k^{-1} \log R(k, k)$. If $a \geq 0$, $d > \log 2$, and

$$\log R(k, \ell) \leq ak + d\ell + o(k) \tag{1.1}$$

uniformly for $1 \leq \ell \leq k$, then

$$\Gamma \leq a + \log 2 + \frac{1}{2} \log(e^d - 1).$$

The numerical application verifies the affine hypothesis (1.1) with $a = 0.544$ and $d = 0.7622$, proved in Proposition 5.3.

We obtain the off-diagonal estimate by iterating the density bounds, with parameters chosen by numerical optimization. The resulting inequalities are verified by exact arithmetic over entire parameter intervals and for both orders of the Ramsey targets.

In Section 2 we prove the maximum-clique argument, which applies to any suitable off-diagonal bound. Sections 3 and 4 give the weighted candidate estimate and the induction that extends it, and in Section 5 we carry out the construction and complete the proof of Theorem 1.1. Section 6 states the formal result and describes its verification. Appendix A contains the interval estimates, and Appendix B describes the numerical data and their verification. Appendix C, which is independent of our proof, examines the HorizonMath claims [15, 8].

2 From off-diagonal to diagonal bounds

We begin with the step that converts an off-diagonal bound into a diagonal one. It is independent of the construction in the later sections.

We work with red–blue colourings of complete graphs, identifying vertex subsets with their induced subgraphs. We write $N_R(v)$ and $N_B(v)$ for the red and blue neighbourhoods of a vertex v . For disjoint vertex sets X, Y , let $e_R(X, Y)$ count the red edges between them, and put $d_R(X, Y) = e_R(X, Y)/(|X||Y|)$ when both sets are nonempty, with $d_R(X, Y) = 0$ otherwise. For a graph G , let $e_R(G)$ count its red edges and let $\omega_R(G)$ be its red clique number, with $\omega_R(\emptyset) = 0$. For nonempty G , write $\delta_R(G)$ for its minimum red degree, and for $|G| \geq 2$ put $d_R(G) = e_R(G)/\binom{|G|}{2}$.

Definition 2.1. We call a function $E : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$ a *valid upper profile* if, for every $\varepsilon > 0$ and all sufficiently large k ,

$$\log R(k, \ell) \leq k(E(\ell/k) + \varepsilon) \quad (1 \leq \ell \leq k).$$

A profile on a subinterval asserts the same uniform bound for ratios in that subinterval.

We first pass to a subgraph of large minimum red degree.

Lemma 2.2. *Let $0 < p < 1$. Every red–blue complete graph G of order $n \geq 1$ has a nonempty induced subgraph W with*

$$\delta_R(W) \geq p(|W| - 1), \quad |W| \geq \frac{2e_R(G)}{n} - pn.$$

Proof. Among the nonempty vertex subsets, choose W maximizing $e_R(W) - p|W|^2/2$. If $|W| \geq 2$ and $v \in W$, comparing W with $W \setminus \{v\}$ gives $|N_R(v) \cap W| \geq p(|W| - \frac{1}{2}) \geq p(|W| - 1)$; for $|W| = 1$ the degree condition is vacuous. Comparing W with the whole vertex set gives $e_R(W) - p|W|^2/2 \geq e_R(G) - pn^2/2$, and $e_R(W) \leq n|W|/2$, which rearranges to the size bound. $\square$

The next observation concerns the common neighbourhoods of the subsets of a largest red clique. The choice of a largest clique bounds the size of a red clique among those neighbours, and a weighted count over all subsets replaces any selection of a single subset.

Lemma 2.3. *Let S be a red clique of maximum size in a red–blue complete graph, with $|S| = M$, and let X be disjoint from S . For $A \subseteq S$, write $X_A = \bigcap_{a \in A} N_R(a) \cap X$, with $X_\emptyset = X$. Then $\omega_R(X_A) \leq M - |A|$, and for every $z \geq 0$,*

$$\sum_{A \subseteq S} z^{|A|} |X_A| = \sum_{x \in X} (1 + z)^{|N_R(x) \cap S|}.$$

Proof. A red clique of order $M - |A| + 1$ in any X_A would join A to form a red clique larger than S . For the identity, both sides count each pair (x, A) with $x \in X$ and $A \subseteq N_R(x) \cap S$ with weight $z^{|A|}$; the right side follows by the binomial theorem. $\square$

We can now convert an affine off-diagonal estimate into a diagonal one.

Theorem 2.4. *Let $a \geq 0$ and $d > \log 2$, and suppose that the affine function $t \mapsto a + dt$ is a valid upper profile on $(0, 1]$. Then*

$$\Gamma \leq a + \log 2 + \frac{1}{2} \log(e^d - 1).$$

Proof. Fix $g > a + \log 2 + \frac{1}{2} \log(e^d - 1)$ and put $\varepsilon = \frac{1}{2}\{g - a - \log 2 - \frac{1}{2} \log(e^d - 1)\}$. It suffices to prove that, for all sufficiently large k , every red–blue complete graph of order at least e^{gk} contains a monochromatic K_k . The profile hypothesis and Ramsey symmetry give, for all sufficiently large k ,

$$R(\ell, k) = R(k, \ell) \leq e^{(a+\varepsilon)k+d\ell} \quad (1 \leq \ell \leq k). \quad (2.1)$$

Suppose, for such a k , some red–blue complete graph G of order $N \geq e^{gk}$ contains no monochromatic K_k . Since $g > \log 2 > 0$, we may assume $k \geq 2$ and $N \geq 32k^3$. After exchanging the colours if necessary, the red density of G is at least $1/2$, so $2e_R(G)/N \geq (N-1)/2$. Apply Lemma 2.2 with $p = \frac{1}{2} - \frac{1}{2k}$: it provides a subgraph W with

$$\delta_R(W) \geq p(|W| - 1), \quad |W| \geq \frac{N}{2k} - \frac{1}{2} \geq \frac{N}{4k}.$$

Let S be a red clique of maximum size in W , and put $M = |S|$ and $X = W \setminus S$. Then $1 \leq M \leq k-1$ and $|X| \geq N/(4k) - k \geq N/(8k) \geq 4k^2$. Each vertex of S has at least $p(|W| - 1) - (M-1) \geq p(|X| - 1) - M$ red neighbours in X . Summing over S and averaging over X ,

$$\frac{1}{|X|} \sum_{x \in X} |N_R(x) \cap S| \geq M \left\{ p - \frac{p+M}{|X|} \right\} \geq Mp - \frac{1}{2} \geq \frac{M}{2} - 1,$$

where the middle step uses $M \leq k$ and $|X| \geq 4k^2$, and the last uses $Mp \geq M/2 - \frac{1}{2}$.

Put $z = e^d - 2$, which is positive because $d > \log 2$. Since X_A contains no blue K_k and $\omega_R(X_A) \leq M - |A|$ by Lemma 2.3, we have $|X_A| < R(M - |A| + 1, k)$, and $M - |A| + 1 \leq k$. Hence, by (2.1),

$$\sum_{A \subseteq S} z^{|A|} |X_A| \leq e^{(a+\varepsilon)k+d} \sum_{j=0}^M \binom{M}{j} z^j e^{d(M-j)} = e^{(a+\varepsilon)k+d} \{2(e^d - 1)\}^M.$$

On the other hand, $u \mapsto (1+z)^u$ is convex and $1+z = e^d - 1 > 1$, so the identity of Lemma 2.3 and the mean degree into S give

$$\sum_{A \subseteq S} z^{|A|} |X_A| = \sum_{x \in X} (e^d - 1)^{|N_R(x) \cap S|} \geq |X| (e^d - 1)^{M/2-1}.$$

Comparing the two displays,

$$|X| \leq e^{(a+\varepsilon)k+d} (e^d - 1) (2\sqrt{e^d - 1})^M \leq e^{(a+\varepsilon)k+d} (e^d - 1) (2\sqrt{e^d - 1})^k,$$

because $2\sqrt{e^d - 1} > 2$ and $M \leq k$. Consequently

$$e^{gk} \leq N \leq 8k|X| \leq 8ke^d (e^d - 1) e^{k\{a+\varepsilon+\log 2+\frac{1}{2}\log(e^d-1)\}} = 8ke^d (e^d - 1) e^{(g-\varepsilon)k},$$

which is a contradiction for all sufficiently large k . Therefore $R(k, k) \leq \lceil e^{gk} \rceil$ for all sufficiently large k , and $\Gamma \leq g$. Let g decrease to the stated value. $\square$

Remark 2.5. Only the affine form of the off-diagonal estimate enters. A bound for a general upper profile follows by applying the theorem to any affine majorant, and, for a concave profile, the best choice is a supporting line. In Section 5, the polygonal profile is dominated by an explicitly chosen affine bound.

3 A weighted form of the candidate lemma

We now turn to the construction of the off-diagonal bound. The starting point is the candidate lemma of Gupta, Ndiaye, Norin and Wei [6]. We allow the sizes of the two candidate sets to enter with different weights.

Definition 3.1. Following [6], a *candidate* is a pair (X, Y) of disjoint nonempty vertex sets. It is (k, ℓ, t) -good if the colouring contains a red K_k in $X \cup Y$, a blue K_t in X , or a blue K_ℓ in Y .

Definition 3.2. The *Ramsey region* $\mathcal{R}$ is the closure, relative to $(0, 1)^2$, of the pairs (x, y) for which $R(k, \ell) \leq x^{-k}y^{-\ell}$ whenever $k + \ell$ is sufficiently large. Following [6], write $\mathcal{R}_*$ for its interior. For a valid upper profile E and $A > 0$, define

$$b_E(A) = \max \left\{ 0, \sup_{0 < t \leq 1} \frac{E(t) - A}{t}, \sup_{0 < t \leq 1} (E(t) - tA) \right\}.$$

Lemma 3.3. If $A, B > 0$ and $B \geq b_E(A)$, then $(e^{-A}, e^{-B}) \in \mathcal{R}$. For every $\delta > 0$, the point $(e^{-A-\delta}, e^{-B-\delta})$ is in $\mathcal{R}_*$.

Proof. The hypothesis gives the two *support inequalities* $A + tB \geq E(t)$ and $B + tA \geq E(t)$ for $0 < t \leq 1$. Use the first when $\ell \leq k$ and the second, with Ramsey symmetry, when $k \leq \ell$. The uniform error is absorbed by arbitrarily small positive additions to A, B . A further addition gives an interior point. $\square$

Theorem 3.4. Fix $\theta > 0$ and $x, y, \mu, p \in (0, 1)$ satisfying

$$(x, y) \in \mathcal{R}_*, \quad x < (1 - \mu)^\theta p^{1/(1-\mu)}.$$

There is L_0 such that, for all positive integers k, ℓ, t with $\ell \geq L_0$, every candidate satisfying

$$d_R(X, Y) \geq p, \quad |X|^\theta |Y| \geq x^{-k} y^{-\ell} \mu^{-\theta t}$$

is (k, ℓ, t) -good.

When $\theta = 1$, this is the candidate lemma of [6, Lemma 12]. The proof follows their density-excess induction, with the size factor $|X||Y|$ replaced by $|X|^\theta |Y|$.

For the application, it is convenient to write $x = e^{-A}$ and $y = e^{-B}$. The strict density condition becomes

$$(1 - \mu)\{A + \theta \log(1 - \mu)\} + \log p > 0. \tag{3.1}$$

If $\ell/k \rightarrow \lambda$, $t/k \rightarrow \alpha$, and $|X| = e^{ku+o(k)}$, $|Y| = e^{kv+o(k)}$, a sufficient strict size condition is

$$\theta u + v > A + \lambda B - \theta \alpha \log \mu. \tag{3.2}$$

Both inequalities must be checked with the same value of θ .

Corollary 3.5. Under the hypotheses of Theorem 3.4, fix $\lambda > 0$ and let

$$z = \frac{A + \lambda B - \theta \lambda \log \mu}{1 + \theta}.$$

For every $g > z$ and all sufficiently large k , a graph on at least e^{gk} vertices with red density at least p contains a red K_k or a blue $K_{\lceil \lambda k \rceil}$.

Proof. A uniformly chosen balanced partition has expected red cross-density equal to the red density of the graph. Hence some balanced partition (X, Y) has $d_R(X, Y) \geq p$. Each part has order at least $e^{gk-O(1)}$, so the size hypothesis of Theorem 3.4 holds with $t = \ell = \lceil \lambda k \rceil$ for sufficiently large k . Any of its three conclusions gives the required clique. $\square$

We now prove Theorem 3.4. The elementary Ramsey recursion

$$R(k, m) \leq R(k-1, m) + R(k, m-1)$$

follows by splitting the neighbours of one vertex according to their colour. Induction with $R(1, m) = R(k, 1) = 1$ therefore gives

$$R(k, m) \leq \binom{k+m-2}{m-1}. \quad (3.3)$$

We also recall the averaging inequality of [6, Lemma 7].

Lemma 3.6. *For disjoint nonempty X, Y ,*

$$\sum_{v \in X} d_R(X, N_R(v) \cap Y) |N_R(v) \cap Y| \geq d_R(X, Y) e_R(X, Y),$$

where an empty neighbourhood contributes zero.

Proof. Write $d_y = |N_R(y) \cap X|$. The left side is $|X|^{-1} \sum_{y \in Y} d_y^2$, which is at least $|X|^{-1} |Y|^{-1} (\sum_y d_y)^2$ by Cauchy–Schwarz. This is the right side. $\square$

The next lemma is a variant of [6, Lemma 9].

Lemma 3.7. *Fix $\alpha \in (0, 1)$ and positive integers k, b . Put $m = \lceil 10\alpha^{-1}b^2 \rceil$. If at least $R(k, m)$ vertices of a set X have blue degree at least $\alpha|X|$ within X , then X contains a red K_k , or a blue clique A of order b with a common blue neighbourhood T satisfying*

$$|T| \geq \frac{1}{2}\alpha^b |X|.$$

Proof. In the absence of a red K_k , the high-degree vertices contain a blue clique S of order m . Summing degrees from S into X gives an average of at least αm neighbours in S . Counting pairs consisting of $x \in X$ and a b -subset of its blue neighbours in S , and using discrete convexity of $u \mapsto \binom{u}{b}$, gives a b -subset $A \subseteq S$ with

$$|T| \geq |X| \frac{\binom{\lfloor \alpha m \rfloor}{b}}{\binom{m}{b}}.$$

The quotient is at least

$$\alpha^b \prod_{i=0}^{b-1} \left(1 - \frac{i+1}{\alpha m}\right) \geq \alpha^b \left(1 - \frac{b(b+1)}{2\alpha m}\right) \geq \frac{1}{2}\alpha^b.$$

All factors are nonnegative by the choice of m . A common blue neighbour cannot belong to A , so T is disjoint from A . $\square$

Proof of Theorem 3.4. Choose perturbed parameters

$$x > x_0, \quad y > y_0, \quad \mu > \mu_0, \quad p < p_0$$

where x_0, y_0, μ_0, p_0 are the parameters in the theorem. The perturbation is small enough that $(x, y) \in \mathcal{R}_*$ and $x < (1 - \mu)^\theta p^{1/(1-\mu)}$ still hold. Since

$$\lim_{\rho \rightarrow \infty} (p^{1/\rho} - \mu)^\rho (1 - \mu)^{\theta - \rho} = (1 - \mu)^\theta p^{1/(1-\mu)},$$

we may choose a fixed $\rho > \max\{1, \theta\}$ with

$$x^{1/\rho} (1 - \mu)^{1-\theta/\rho} + \mu < p^{1/\rho}. \quad (3.4)$$

The limit follows by expanding $p^{1/\rho} = 1 + (\log p)/\rho + O(\rho^{-2})$ and taking logarithms. Also $x^{1/\theta} + \mu < 1$. Set

$$f(\beta) = x^{1/\rho} (1 - \beta)^{1-\theta/\rho} + \mu^{\theta/\rho} \beta^{1-\theta/\rho}.$$

Its derivative is positive before $\beta = \mu/(\mu + x^{1/\theta}) > \mu$. Thus its maximum on $[0, \mu]$ is $f(\mu)$, the left side of (3.4). Choose a sufficiently small fixed $\varepsilon > 0$ so that

$$(x + \varepsilon, y + \varepsilon) \in \mathcal{R}_*, \quad p - \varepsilon > 0, \quad \mu + \varepsilon < 1, \quad \max_{0 \leq \beta \leq \mu + \varepsilon} f(\beta) < (p - \varepsilon)^{1/\rho}. \quad (3.5)$$

For $s = k + t$, let $\delta_s = \varepsilon/s$. We prove, by induction on s , the stronger assertion that a candidate is (k, ℓ, t) -good whenever

$$z = d_R(X, Y) - p + \delta_s > 0, \quad Q = z^\rho |X|^\theta |Y| \geq x^{-k} y^{-\ell} \mu^{-\theta t}, \quad (3.6)$$

uniformly for sufficiently large ℓ . The cases $k = 1$ or $t = 1$ are immediate.

First delete from X any vertex with red degree less than $(p - \delta_s)|Y|$ into Y . To see that Q does not decrease, write

$$Q = \{e_R(X, Y) - (p - \delta_s)|X||Y|\}^\rho |X|^{\theta - \rho} |Y|^{1 - \rho}.$$

The expression in braces is positive and increases on deletion, while $\theta - \rho < 0$. The process cannot delete every vertex, since the excess-edge expression would then be zero. Rename the remaining set X . We now have

$$|N_R(v) \cap Y| \geq (p - \delta_s)|Y| \quad (v \in X). \quad (3.7)$$

An interior region point gives an eventual Ramsey bound without a further error term: indeed, choose an eventual-bound point slightly above it in both coordinates using the definition of $\mathcal{R}$ and its coordinatewise downward closure. Therefore, if

$$|Y| \geq (x + \varepsilon)^{-k} (y + \varepsilon)^{-\ell},$$

the Ramsey bound inside Y proves the assertion for sufficiently large ℓ . Otherwise (3.6) and $z \leq 1 + \varepsilon$ imply

$$|X|^\theta > (1 + \varepsilon)^{-\rho} \left(\frac{x + \varepsilon}{x} \right)^k \left(\frac{y + \varepsilon}{y} \right)^\ell \mu^{-\theta t}. \quad (3.8)$$

In particular there is a fixed $c > 0$ such that $|X| \geq e^{c(k+\ell+t)}$ for all sufficiently large ℓ .

Choose an integer $b = b(s) = O(\log(s+2))$, with a sufficiently large implied constant, so that

$$\left(\frac{\varepsilon}{(1+\varepsilon)s^2}\right)^\rho 2^{-\theta} \left(\frac{\mu+\varepsilon}{\mu}\right)^{\theta b} \geq 1. \quad (3.9)$$

Let $m = \lceil 10(\mu+\varepsilon)^{-1}b^2 \rceil$ and let W be the vertices of X of blue degree at least $(\mu+\varepsilon)|X|$. By (3.3),

$$\log R(k, m) = O(\log^3(s+2)).$$

In view of (3.8), $R(k, m)/|X|$ is smaller than any fixed negative power of $s+2$, uniformly in s , once ℓ is sufficiently large. This follows because $C \log^3(s+2) - cs$ is bounded above for $s \geq 2$, and the additional factor $e^{-c\ell}$ tends to zero.

If $|W| \geq R(k, m)$, Lemma 3.7 gives the required red clique or a blue book (A, T) with $|A| = b$ and $|T| \geq (\mu+\varepsilon)^b |X|/2$. If $b \geq t$, then A contains the required blue clique. Otherwise, the degree bound into the unchanged set Y gives

$$d_R(T, Y) - p + \delta_{s-b} \geq \delta_{s-b} - \delta_s \geq \varepsilon/s^2.$$

Combining this with $Q \leq (1+\varepsilon)^\rho |X|^\theta |Y|$ and (3.9) yields

$$[d_R(T, Y) - p + \delta_{s-b}]^\rho |T|^\theta |Y| \geq \mu^{\theta b} Q \geq x^{-k} y^{-\ell} \mu^{-\theta(t-b)}.$$

Apply the induction hypothesis to (T, Y) . A blue K_{t-b} in T joins A to give a blue K_t in X . A red K_k in $T \cup Y$ or a blue K_ℓ in Y already gives the required conclusion.

We may assume $|W| < R(k, m)$. Put $Y_v = N_R(v) \cap Y$. Lemma 3.6, followed by deletion of the terms with $v \in W$, gives

$$\sum_{v \notin W} d_R(X, Y_v) |Y_v| \geq d_R(X, Y) e_R(X, Y) - |W| |Y|.$$

Since $e_R(X, Y) \geq (p-\varepsilon)|X||Y|$, the uniform estimate on $|W|/|X|$ makes the last loss at most $\varepsilon e_R(X, Y)/s^3$. The weights on $v \notin W$ sum to at most $e_R(X, Y)$. Hence there is $v \notin W$ for which, with $Y' = Y_v$,

$$d_R(X, Y') \geq d_R(X, Y) - \varepsilon/s^3. \quad (3.10)$$

Set

$$\begin{aligned} X_R &= N_R(v) \cap X, & X_B &= N_B(v) \cap X, \\ q &= |Y'|/|Y|, & \gamma &= |X_R|/|X|, & \beta &= |X_B|/|X|. \end{aligned}$$

Then $q \geq p - \varepsilon$, $\beta \leq \mu + \varepsilon$, and $\gamma + \beta = 1 - |X|^{-1}$. Write $z_R = d_R(X_R, Y') - p + \delta_{s-1}$ and define z_B similarly when the children are nonempty.

If (X_R, Y') satisfies the induction hypothesis with targets $(k-1, \ell, t)$, a red K_{k-1} extends by v , and either blue conclusion suffices directly. Similarly, if (X_B, Y') satisfies it with targets $(k, \ell, t-1)$, a blue K_{t-1} in X_B extends by v , and the other two conclusions suffice. We may therefore assume that neither pair satisfies the induction hypothesis. Let $[w]_+ = \max\{w, 0\}$. The positive parts of their excesses then satisfy

$$[z_R]_+ \leq z x^{1/\rho} \gamma^{-\theta/\rho} q^{-1/\rho}, \quad [z_B]_+ \leq z \mu^{\theta/\rho} \beta^{-\theta/\rho} q^{-1/\rho}.$$

An empty set X_R or X_B contributes zero to the next calculation. Every edge from v to Y' is red, so

$$\begin{aligned} d_R(X, Y') - p + \delta_{s-1} &\leq \gamma[z_R]_+ + \beta[z_B]_+ + \frac{1 + \varepsilon}{|X|} \\ &\leq zq^{-1/\rho}f(\beta) + \frac{1 + \varepsilon}{|X|} \leq z + \frac{1 + \varepsilon}{|X|}, \end{aligned}$$

where we used $\gamma \leq 1 - \beta$, $\rho > \theta$, and (3.5). On the other hand, (3.10) gives

$$d_R(X, Y') - p + \delta_{s-1} \geq z + \delta_{s-1} - \delta_s - \varepsilon/s^3 \geq z + \varepsilon/s^2 - \varepsilon/s^3.$$

These inequalities contradict (3.8) for sufficiently large ℓ , uniformly in s . This completes the stronger induction.

Finally, a candidate in the original theorem has excess at least $p_0 - p$ for the perturbed parameters. The ratio of its guaranteed potential to the threshold in (3.6) is at least

$$(p_0 - p)^\rho (x/x_0)^k (y/y_0)^\ell (\mu/\mu_0)^{\theta t},$$

which is at least one for all sufficiently large ℓ . The stronger assertion applies and proves the theorem. $\square$

4 Extending the candidate lemma

We wish to apply an estimate for (k, ℓ, t) -good candidates at larger values of t . The difficulty is to retain the information supplied by Y as we pass to smaller subsets of X . A minimum degree condition into Y is enough. Such minimum cross-degree conditions also appear in the multicolour book argument of Balister et al. [1]. Here we use the condition to extend a candidate estimate, with previously proved density bounds handling the red-dense case.

Lemma 4.1. *If $d_R(X, Y) \geq p_0 > p_*$, then*

$$X_* := \{x \in X : |N_R(x) \cap Y| \geq p_*|Y|\} \quad \text{satisfies} \quad |X_*| \geq \frac{p_0 - p_*}{1 - p_*}|X|.$$

Every nonempty $T \subseteq X_$ satisfies $d_R(T, Y) \geq p_*$.*

Proof. The number of red cross-edges is at most

$$|X_*||Y| + p_*(|X| - |X_*|)|Y|.$$

Compare with $p_0|X||Y|$ and rearrange. Every vertex of a subset $T \subseteq X_*$ still has at least $p_*|Y|$ red neighbours in Y . Summing these degree bounds gives the last assertion. $\square$

We first state the induction for integer clique sizes.

Lemma 4.2. *Fix k, ℓ , integers $1 \leq t_0 \leq t_1$, a positive integer B_Y , and $p_* \in (0, 1)$. Let A_t be positive integer thresholds. Suppose:*

1. *Every candidate with $|X| \geq A_{t_0}$, $|Y| \geq B_Y$, and red degree at least $p_*|Y|$ from every vertex of X into Y is (k, ℓ, t_0) -good.*
2. *For each $t_0 < t \leq t_1$, a previously proved density bound with threshold $q_t \in (0, 1)$ gives a red K_k or a blue K_t in every graph of order at least A_t and red density at least q_t .*

3. $A_{t-1} \leq (1 - q_t)(A_t - 1)$ for each $t_0 < t \leq t_1$.

Then the first assertion holds with (t_0, A_{t_0}) replaced by (t, A_t) for every $t_0 \leq t \leq t_1$.

Proof. We induct on t , the initial case being the first hypothesis. Suppose $t > t_0$. If the red density within X is at least q_t , the second hypothesis gives a red K_k or a blue K_t in X . Otherwise some $x \in X$ has a blue neighbourhood $T \subseteq X$ of order at least $(1 - q_t)(|X| - 1) \geq A_{t-1}$. Every vertex of T satisfies the same red degree bound into Y , so the induction hypothesis applies to (T, Y) . A red K_k in $T \cup Y$ or a blue K_ℓ in Y proves the assertion. In the remaining case, a blue K_{t-1} in T extends by x to a blue K_t in X . $\square$

To use the preceding lemma with piecewise-defined exponent functions, we need density estimates that allow a small increase in the blue target. We record that requirement in the following notation.

Definition 4.3. For $r, z > 0$ and $q \in (0, 1)$, write $D(r, z, q)$ if there exists $\delta > 0$ such that, for every $\varepsilon > 0$ and all sufficiently large k , every graph on at least $e^{k(z+\varepsilon)}$ vertices with red density at least q contains a red K_k or a blue $K_{\lceil (r+\delta)k \rceil}$.

By monotonicity, a statement $D(r, z, q)$ also covers smaller blue targets. The margin δ makes the statement robust under a small increase in the blue target, allowing its use on both sides of a nearby junction in a piecewise-defined function. Unlike a valid upper profile, this statement assumes a lower bound on the red density.

Theorem 4.4. Fix $0 < \alpha < \lambda$, $v > 0$, and $0 < p_* < p_0 < 1$. Let u be positive, increasing, continuous and piecewise C^1 on $[\alpha, \lambda]$. Suppose there are seed parameters $A_0, B_0, \theta_0 > 0$, $\mu_0 \in (0, 1)$, with $(e^{-A_0}, e^{-B_0}) \in \mathcal{R}_*$, such that

$$(1 - \mu_0)\{A_0 + \theta_0 \log(1 - \mu_0)\} + \log p_* > 0, \quad (4.1)$$

$$\theta_0 u(\alpha) + v > A_0 + \lambda B_0 - \theta_0 \alpha \log \mu_0. \quad (4.2)$$

On each closed smooth piece I , suppose fixed $z_I > 0$ and $q_I \in (0, 1)$ are supplied by an earlier robust density statement $D(\sup I, z_I, q_I)$ and satisfy

$$u(s) > z_I, \quad u'(s) > -\log(1 - q_I) \quad (s \in I), \quad (4.3)$$

using one-sided derivatives at endpoints. Then, for every $\varepsilon > 0$ and all sufficiently large k , every candidate with

$$d_R(X, Y) \geq p_0, \quad |X| \geq e^{k(u(\lambda)+\varepsilon)}, \quad |Y| \geq e^{k(v+\varepsilon)}$$

is $(k, \lceil \lambda k \rceil, \lceil \lambda k \rceil)$ -good. If $u(\lambda) < v$, the same data establish $D(\lambda, v, p_0)$.

Proof. Fix $\varepsilon > 0$. There are finitely many pieces, and all inequalities in the hypotheses are strict on their closed domains. On the interior of a piece its density statement is valid for the current target and has positive size and slope margins. At a junction, choose the density statement from the side with the smaller one-sided derivative. Both adjacent limiting derivatives exceed $-\log(1 - q)$ for that statement, and its size inequality persists by continuity of u . If the statement is taken from the left, its positive margin in the blue target allows its use a little to the right. If it is taken from the right, monotonicity allows its use to the left.

It follows that $[\alpha, \lambda]$ has a finite open cover by neighbourhoods with fixed density controls and a common positive margin η such that, throughout each neighbourhood,

$$u(s) \geq z + \eta, \quad u'(s) \geq -\log(1 - q) + \eta$$

where the derivative exists. Choose k large enough that all of these finitely many density statements apply. Extend u a short distance beyond each endpoint, affinely with its endpoint derivative, when integer rounding requires it; the endpoint conditions persist.

Apply Lemma 4.1 once to obtain the minimum degree condition into Y . This loses only a fixed proportion of X , which is absorbed by the εk margin. Set $t_0 = \lceil \alpha k \rceil$, $t_1 = \lceil \lambda k \rceil$, $\ell = \lceil \lambda k \rceil$, and use the integer thresholds

$$A_t = \lceil \exp(k[u(t/k) + \varepsilon/3]) \rceil, \quad B_Y = \lceil \exp(k[v + \varepsilon/3]) \rceil.$$

By (4.2), the logarithm of $A_{t_0}^{\theta_0} B_Y$ exceeds

$$A_0 k + B_0 \ell - \theta_0 t_0 \log \mu_0$$

for large k . The error from rounding αk and λk is $O(1)$. Equation (4.1) and Theorem 3.4 give the initial case of Lemma 4.2.

For every $t_0 < t \leq t_1$, the interval $[(t-1)/k, t/k]$ lies in one of the finitely many neighbourhoods when k is large. Use its controls (z, q) . The density statement applies at A_t because $u(t/k) > z$. Integration of the slope inequality gives

$$k\{u(t/k) - u((t-1)/k)\} \geq -\log(1-q) + \eta.$$

Consequently,

$$\frac{A_{t-1}}{A_t - 1} \leq e^{-\eta}(1-q) + o(1) < 1-q,$$

uniformly in t . Here the rounding error is uniform because u has a positive minimum. The hypotheses of Lemma 4.2 now hold. The sets remaining after the deletion have orders at least A_{t_1} and B_Y , respectively, proving the first assertion.

For the last assertion, choose $\lambda' > \lambda$ sufficiently close to λ and extend the last piece affinely. Its density input is valid at λ' by robustness. The strict endpoint, slope, and seed conditions survive, including $u(\lambda') < v$. A graph on $e^{(v+\varepsilon)k}$ vertices with red density at least p_0 has a balanced partition with red cross-density at least p_0 . Both parts have exponent $v + \varepsilon + o(1)$. Applying the first assertion at λ' proves $D(\lambda, v, p_0)$. $\square$

In later applications, we may replace a weighted density bound by a statement $D(r, z, q)$ proved by an earlier application of the theorem. The initial candidate estimate still uses both X and Y ; only the density bound for the red-dense case is replaced.

Proposition 4.5. *Keep the hypotheses on u and the density estimates in Theorem 4.4, and suppose a continuous valid upper profile E is available near α . Replace the weighted seed hypotheses by $u(\alpha) > E(\alpha)$. Then*

$$\log R(k, \lceil \lambda k \rceil) \leq ku(\lambda) + o(k).$$

For every $v > u(\lambda)$, the same data give $D(\lambda, v, p_0)$ for any fixed $p_0 \in (0, 1)$.

Proof. Use the same thresholds A_t and the same neighbourhood cover as in the preceding proof. The unconditional input gives $A_{t_0} \geq R(k, t_0)$ for large k . Induct on t : the dense branch gives a red K_k or a blue K_t directly, and the other branch has a blue neighbourhood with at least A_{t-1} vertices. A blue K_{t-1} supplied by the induction extends by the chosen vertex. This argument does not use a second set Y . The result holds with arbitrarily small positive exponential slack. For the robust conclusion, extend the final piece slightly while retaining $u(\lambda') < v$ and all density conditions. The resulting unconditional conclusion implies the stated density assertion. $\square$

5 Proof of Theorem 1.1

We now assemble the off-diagonal estimate needed in Theorem 2.4. The first step is to convert the density statements of the preceding section into an unconditional Ramsey bound. As before, we induct on the blue target, using a density statement when the graph is red-dense and a blue neighbourhood otherwise.

Theorem 5.1. *Let $0 < L = \lambda_0 < \lambda_1 < \dots < \lambda_n = 1$. Suppose a continuous valid input profile E_0 is available near L . For each i , suppose $D(\lambda_i, v_i, p_i)$ has already been proved. Let F be continuous and affine with positive slope s_i on $[\lambda_{i-1}, \lambda_i]$, and assume*

$$F(L) > E_0(L), \quad F(\lambda_{i-1}) > v_i, \quad s_i > -\log(1 - p_i).$$

Then, uniformly for $L \leq \ell/k \leq 1$,

$$\log R(k, \ell) \leq kF(\ell/k) + o(k).$$

Proof. On the interior of interval i , the statement $D(\lambda_i, v_i, p_i)$ applies to all current blue targets up to that interval's right endpoint. Since F is increasing, the size inequality holds throughout the interval. At a knot choose the controls of the adjacent interval with smaller slope. Both adjacent slopes exceed the cost of those controls. Their size margin extends across the knot by continuity; their target margin extends by robustness when the controls are taken from the left. Thus, as in the proof of Theorem 4.4, there is a finite neighbourhood cover with uniform positive size and slope margins.

Fix $\varepsilon > 0$ and set $A_\ell = \lceil e^{k(F(\ell/k) + \varepsilon)} \rceil$, making a continuous affine extension over the $O(k^{-1})$ endpoint adjustment if needed. At $\ell_0 = \lceil Lk \rceil$, the strict input inequality and continuity give $A_{\ell_0} \geq R(k, \ell_0)$ for large k . On each subsequent step select controls covering $[(\ell - 1)/k, \ell/k]$. They give $A_{\ell-1} \leq (1 - p)(A_\ell - 1)$ by integration of the strict slope bound. If the graph at order A_ℓ has red density at least p , its density statement gives the conclusion. Otherwise a vertex has a blue neighbourhood of size at least $A_{\ell-1}$, to which induction applies. This proves the asserted bound uniformly; let ε decrease to zero. $\square$

Proposition 5.2. *Consider a finite list of profile, region, candidate, and robust density statements, starting from established Ramsey estimates. Suppose every subsequent statement follows from earlier statements by Lemma 3.3, Theorems 3.4, 4.4 and 5.1, Proposition 4.5, taking a pointwise minimum of established upper bounds, or pointwise enlargement of an upper bound. If the hypotheses of each result used are satisfied, every statement in the list is valid.*

In particular, if a profile F obtained from Theorem 5.1 is replaced by a continuous piecewise-affine majorant P , checking $P \geq F$ at every knot of their common subdivision suffices to justify reuse of P as an input profile on that interval. The support check must use a valid bound on the whole ratio domain $(0, 1]$.

Proof. Induct along the finite list. For the last assertion, $P - F$ is affine between consecutive knots. Combine the interval bound with the previously established bounds on the rest of the ratio domain before applying Lemma 3.3. $\square$

We take the initial input from [6, Theorem 1]:

$$U(t) = (1 + t) \log(1 + t) - t \log t + \left(-\frac{t}{4} + \frac{3t^2}{100} + \frac{2t^3}{25} \right) e^{-t}. \quad (5.1)$$

In the notation of version 2 of that theorem,

$$\log R(k, \ell) \leq kU(\ell/k) + o(k) \quad (1 \leq \ell \leq k),$$

with an error uniform in ℓ . This is the only external Ramsey estimate used in the construction.

In the numerical construction, the functions u of Theorem 4.4 are piecewise affine; we refer to their graphs as paths. The construction applies the preceding rules in eight rounds: five with mesh denominator 2000, followed by three with mesh denominator 6000. Its data comprise about 1.7 million density records, almost all with weighted starts, and about 814 million references to earlier density statements along these paths. Appendix B describes the records, their compressed representation, and the exact counts. Decimal constants are exact rationals throughout; in particular $0.544 = 68/125$, $0.7622 = 3811/5000$ and $3.68395 = 73679/20000$.

Proposition 5.3. *The affine function $t \mapsto 0.544 + 0.7622t$ is a continuous valid upper profile on $(0, 1]$.*

Proof. We apply the construction in Appendix B, starting from U . Its initial stage replaces U by a polygonal majorant, whose validity follows from tangent estimates. Each subsequent density statement is obtained in one of two ways. A weighted initial estimate, together with the recorded size and slope inequalities, gives an application of Theorem 4.4. An unconditional initial estimate instead gives an application of Proposition 4.5. Every density statement used in either argument precedes the statement being proved. Thus all of them follow by induction along the finite list.

Theorem 5.1 gives off-diagonal upper profiles in each of the eight rounds. We take pointwise minima and then concave majorants; these remain valid upper bounds by Proposition 5.2.

The required inequalities are checked as described in Section 6 and Appendix B. They include the support inequalities, the estimates near ratio zero from Appendix A, and the domination of the reconstructed polygon by the line $0.544 + 0.7622t$ at every knot; between consecutive knots the difference is affine, so the line dominates the polygon throughout its range, and it is a valid upper profile there by Proposition 5.2. Below the first knot L of the polygon, the comparison with U in Appendix A extends the domination to the remaining small ratios, using the checked slope inequality $0.7622 < U'(L)$. $\square$

Proof of Theorem 1.1. Rational enclosures give $0.7622 > \log 2$. Proposition 5.3 and Theorem 2.4 imply

$$\Gamma \leq 0.544 + \log 2 + \frac{1}{2} \log(e^{0.7622} - 1) = \log(2e^{0.544} \sqrt{e^{0.7622} - 1}).$$

The terminal calculation compares squares, avoiding logarithms and square roots. Twelve-term enclosures of the two exponentials give

$$e^{0.544} \leq \frac{17228847}{10^7}, \quad e^{0.7622} \leq \frac{21429857}{10^7}, \quad (5.2)$$

and exact rational arithmetic gives

$$4 \left(\frac{17228847}{10^7} \right)^2 \left(\frac{21429857}{10^7} - 1 \right) < 3.68395^2.$$

Hence $2e^{0.544} \sqrt{e^{0.7622} - 1} < 3.68395$, so $\Gamma < \log 3.68395$, which proves the theorem for all sufficiently large k . $\square$

Remark 5.4. Theorem 2.4 gives the final improvement: the affine bound evaluated at $t = 1$ gives only $\Gamma \leq 1.3062$, that is, the diagonal base $e^{1.3062} = 3.6921 \dots$, while Theorem 2.4 gives $3.68388 \dots$. The iteration was stopped after eight rounds; the constant reflects the chosen verification budget.

6 The formal verification

The Lean development proves the inequalities of Section 5, the soundness of the finite checks, and the graph-theoretic results. This section states the formal result and describes how the numerical inequalities are checked; the components and statistics are listed in Appendix B.

6.1 The formal statement

The concluding theorem is stated in the file `RamseyBelow3684.lean` of the repository [9]:

$$\begin{array}{l} \text{Compact3684.ramsey_le_368395 :} \\ \forall^f k \in \mathbb{N}, \quad \text{ramseyNumber } k \leq (73679/20000)^k, \end{array}$$

where $\forall^f$ is Lean’s eventual quantifier along the filter `atTop`, so that the statement holds for all sufficiently large k , with no explicit threshold extracted, and the inequality is between real numbers. Since $73679/20000 = 3.68395$ exactly, this is Theorem 1.1. The definition `ramseyNumber` is taken unchanged from the formalization [7]. The command `#print axioms` reports exactly `propext`, `Classical.choice` and `Quot.sound`, the standard classical axioms of Lean’s logic. The proof contains no `sorry` or additional axioms. The numerical checks do not rely on `native_decide`; they are evaluated by the kernel.

The graph-theoretic inputs are formalized in full: Theorem 3.4, in a uniform, self-contained version; the induction of Lemma 4.2; the off-diagonal estimate of Theorem 5.1, together with a monotone extension from the red-target grid to all integer targets; and the finite-graph inequality proved in Section 2. The induction is formalized for the grid used in the records; the piecewise- C^1 statement of Theorem 4.4 is its mathematical generalization. The soundness of the record and path checkers—the statement that acceptance of the finite data implies the corresponding density and profile statements—is itself a Lean theorem, proved by well-founded induction on the reference order described in Appendix B.

6.2 Verifying the numerical inequalities

Checked naively, the construction requires several elementary-function enclosures for each of its 1.7 million records, and kernel arithmetic at that volume is prohibitive. We instead spend a small part of the construction’s numerical slack to make the enclosures shareable. Fix $\varepsilon = 18 \cdot 10^{-6}$ and $h = 16 \cdot 10^{-6}$, and transform every statement by

$$E(t) \mapsto E(t) + \varepsilon t, \quad v \mapsto v + \varepsilon r, \quad B \mapsto B + \varepsilon, \quad c \mapsto c + \varepsilon,$$

with A unchanged. Here v and r are the exponent and target ratio of a record, p is its density threshold, and c is its blue-neighbourhood cost, satisfying $c > -\log(1 - p)$, as in Appendix B. We then round μ and $1 - p$ downward onto a rational grid whose consecutive ratios are at most $1 + h$. Since $t \leq 1$, both support inequalities remain valid: $A + t(B + \varepsilon) \geq E(t) + \varepsilon t$ and $B + \varepsilon + tA \geq E(t) + \varepsilon t$. The gained ε pays for both roundings, and the strict density floor moves by less than the recorded margins. These scalar implications are Lean lemmas with explicit hypotheses.

An identity on the grid makes the accounting exact: under this transformation, every interval size margin in a path or an application of Theorem 5.1 decreases by exactly ε/N , where N is the mesh denominator—not by the path length times ε/N . Before the transformation, all these recorded margins exceed 10^{-8} , whereas $\varepsilon/2000 = 9 \cdot 10^{-9}$ and $\varepsilon/6000 = 3 \cdot 10^{-9}$. The cancellation behind this claim is proved once, as the lemma `linearLift.compose`, and applied to every path.

After the transformation, the controls take 351,607 distinct logarithm arguments in place of several million. Each is enclosed once, to eighteen decimal places, by the dyadically reduced series of Appendix A with twenty terms, and checked by the kernel. A further 20,000 kernel-checked tangent bounds certify the initial polygonal majorant against the formalized input theorem, and 126,192 checks cover the support and profile inequalities.

All eight round assemblies, the final theorem, and a recheck of the final theorem against the pinned upstream sources were compiled and kernel-checked. The complete observed verification took 56 hours 43 minutes, and the final recheck peaked at 5.40 GiB of sampled resident memory. The repository [9] records the run reports, the reuse of cached upstream builds, and the source audit of the pinned upstream; Appendix B lists the components and statistics.

AI Usage Disclosure

OpenAI’s GPT-5.6 Sol was used for exploratory discussions during the development of the proofs. OpenAI’s GPT-6 Astra carried out the numerical construction, its control quantization, and the development and execution of the Lean verification, and assisted with drafting. Anthropic’s Claude Fable 5 assisted with drafting and revising the manuscript and with preparing the released repository. The author takes full responsibility for the paper’s contents and correctness.

A Verification on parameter intervals

This appendix explains how to check the analytic hypotheses of Section 4 on whole intervals. The paths are piecewise affine, so their size conditions reduce to endpoint comparisons.

Proposition A.1. *Let x be the value of an increasing piecewise-affine path at its right endpoint. Number the intervals from right to left, with lengths $h_1, \dots, h_m > 0$ and slopes $s_1, \dots, s_m > 0$. On interval j , use an earlier robust density statement with exponent v_j and threshold q_j . Put $C_j = \sum_{i=1}^j h_i s_i$.*

The size inequalities on all intervals are equivalent to

$$x > \max_{1 \leq j \leq m} (v_j + C_j).$$

The slope inequalities are $s_j + \log(1 - q_j) > 0$. The exponent at the initial endpoint is exactly $x - C_m$.

Proof. The left endpoint of interval j has value $x - C_j$. As the path increases, this is its smallest value on that interval. Subtraction proves the size equivalence, and the other assertions are the definitions of the slope and accumulated increment. $\square$

Lemma A.2. *On a positive interval where E is piecewise affine, it suffices to check both support inequalities at every knot. If $E = U$ on $(0, L]$ and*

$$U(L) - LU'(L) \leq A \leq U'(L),$$

the two support suprema over $(0, L]$ are attained at L . Thus the same knot tests cover that small-ratio interval as well.

Proof. On an affine piece, each of $A + tB - E(t)$ and $B + tA - E(t)$ is affine, so its minimum occurs at an endpoint. For the small-ratio statement, U is concave. To verify this directly, differentiate (5.1):

$$U''(t) = -\frac{1}{t(1+t)} + e^{-t} (0.56 + 0.11t - 0.45t^2 + 0.08t^3) < 0 \quad (0 < t \leq 1).$$

The polynomial is at most 0.75, and $0.75t(1+t)e^{-t} \leq 1.5/e < 1$. Consequently $C(t) = U(t) - tU'(t)$ is increasing and U' is decreasing. The derivatives of the two support expressions are

$$\frac{d}{dt} \frac{U(t) - A}{t} = \frac{A - C(t)}{t^2} \geq 0, \quad \frac{d}{dt} \{U(t) - tA\} = U'(t) - A \geq 0.$$

This proves the endpoint assertion. □

Extension below the first knot. Some unconditional starting records use ratios below the first polygonal knot L . Let P have first slope s , and extend its first segment linearly to $0 < t < L$. The verification checks $P(L) > U(L)$ and $s < U'(L)$. Concavity of U therefore gives

$$P(L) + s(t - L) > U(L) + U'(L)(t - L) \geq U(t) \quad (0 < t < L).$$

Thus this extension is itself a valid upper bound. In particular, an unconditional start below L does not require extrapolating an unverified numerical approximation to U . The terminal line of Proposition 5.3 is handled by the same comparison: its value at L dominates $P(L) > U(L)$ by the knot checks, and its slope satisfies $0.7622 < U'(L)$, so the displayed inequality extends its domination over $(0, L)$ as well. At the other endpoint, Ramsey symmetry supplies the continuous extension $tP(1/t)$ for $t > 1$ close to 1. Thus the right-hand target margins required for unconditional records at ratio 1 are also available.

If the actual input is a pointwise minimum of several polygons, its subdivision includes all rational intersections of their affine pieces. A subsequent concave majorant is checked on this full subdivision. Concavity can accelerate searches for the support maxima, but the finite knot inequalities establish their validity directly. When the tangent-window condition in Lemma A.2 is unavailable, the small-ratio inequalities require their own analytic or interval justification.

Bounds for logarithms and exponentials. For rational $z > 0$, write $z = 2^a y$ with $a \in \mathbb{Z}$ and $1 \leq y \leq 2$, put $w = (y - 1)/(y + 1)$, and, for $n \geq 1$, use

$$\log y = 2 \sum_{j=0}^{n-1} \frac{w^{2j+1}}{2j+1} + \mathcal{E}_n, \quad 0 \leq \mathcal{E}_n \leq \frac{2w^{2n+1}}{(2n+1)(1-w^2)}.$$

Here $0 \leq w \leq 1/3$; the same formula at $y = 2$ encloses $\log 2$. For $0 \leq z \leq 2$, the exponential series through degree n has a positive tail at most $2z^{n+1}/(n+1)!$ when $n \geq 2$. Reciprocal intervals enclose e^{-z} . Every operation is rounded outward on a rational grid. These formulas justify the elementary-function comparisons without assuming the correctness of a floating-point optimizer.

B The numerical data and their verification

B.1 The records

The proof is organized as a finite list of records. A density record has a target ratio r , exponent v , density floor p , and a blue-neighbourhood cost c with

$$c + \log(1 - p) > 0.$$

It asserts $D(r, v, p)$. A record also carries its justification and references only previously established records. There are two principal justifications. A weighted record specifies a path for the induction

of Theorem 4.4, the initial parameters (A, B, θ, μ, p_*) , and the exponent v . An unconditional record instead starts from a previously established unconditional bound. The strict endpoint gap $u(r) < v$ makes each output robust, as in Theorem 4.4 and Proposition 4.5. A path with no intervals uses its initial estimate directly. For a weighted start, $u(r) < v$ and the strict initial size condition put v strictly above the threshold in Corollary 3.5; a small increase of both blue targets preserves the inequalities. An unconditional start uses the input profile near its target, or the target margin already established for an earlier unconditional record.

For a piecewise-affine path u from ratio α to $r = i/N$, the data list the earlier density references from right to left. The j th reference supplies an exponent v_j and a slope c_j over an interval of width $1/N$. The value at the right endpoint is $x = u(r) = v - g$, where $g > 0$; the other path values are reconstructed by subtracting the slope increments. If the path has m intervals, put $C_j = N^{-1} \sum_{h=1}^j c_h$. The exact checks are

$$x > v_j + C_j \quad \text{for every referenced interval,} \quad u(\alpha) = x - C_m > 0.$$

These are precisely the path-size conditions in Proposition A.1. For a weighted start, the remaining checks are the two support inequalities from Lemma 3.3, the strict seed-density inequality (4.1), and the strict seed-budget inequality (4.2). For an unconditional start, the check is $u(\alpha) > E(\alpha)$ or the corresponding previously proved unconditional record.

The checker verifies both the numerical inequalities and the logical types of the references. In particular, a conditional density statement cannot be used as an unconditional starting bound. References must have the required target ratio and must precede the current record. In each round, records are arranged in rows indexed by the target ratio; a column number distinguishes the density estimates at each row. The records are ordered lexicographically by round, row, and column. Within a round, references along a path have smaller row, or the same row and smaller column. This order permits induction over all the records.

Consecutive path references with the same column number are stored as a single *constant-column run*, recording the column and the run length. Successive references in a run have decreasing, consecutive row indices.

B.2 The resulting upper bound

For each profile obtained from Theorem 5.1, the data record an initial ratio and exponent, followed by one density reference per interval. The checker verifies the initial unconditional bound, the interval size inequalities, and the blue costs in that theorem. It reconstructs the profile by exact rational summation. Between rounds, pointwise minima include all exact affine intersections, and any subsequent concave majorant is checked to dominate the minimum at its knots. The resulting profile is therefore available to the next round by Proposition 5.2.

The full data comprise eight rounds: five with mesh denominator 2000 and three with mesh denominator 6000. There are 1,695,911 density records (1,691,788 weighted, 4,123 unconditional) and 813,922,863 density references along these paths. The references are represented by 41,793,102 constant-column runs, subdivided into 41,794,061 runs for checking. The terminal inference checks the domination of the reconstructed polygon by the line $0.544 + 0.7622t$ at every knot, and the tail comparison of Appendix A below the first knot; between consecutive knots the difference is affine. After the slack transformation of Section 6, the reconstructed final profile is dominated by the line. The replay also selects a supporting line of that profile and applies the transfer to it, reporting the upper bound 3.683749312877; the chosen line $0.544 + 0.7622t$ gives 3.683889327713. The terminal enclosures (5.2) are obtained from the rational series of Appendix A.

Table 1: The scale of the formal proof.

Rounds (mesh 2000 / mesh 6000)	5 + 3
Density records (weighted)	1,695,911 (1,691,788)
Density references along paths	813,922,863
Constant-column runs (split for checking)	41,793,102 (41,794,061)
Shared logarithm enclosures	351,607
Initial tangent bounds	20,000
Support and profile checks	126,192
Transfer over the selected supporting line	3.683749312877
Proved base	3.68395
Generated Lean source	≈ 2.6 GB

B.3 The Lean development

The development uses Lean 4 (toolchain 4.32.1) and Mathlib [10]. It imports the definition of the Ramsey number and the input theorem for the profile (5.1) from the formalization [7], in the copy distributed with [12]. Its reusable theorems are the uniform weighted candidate theorem, the induction of Lemma 4.2, the off-diagonal estimate of Theorem 5.1 with monotone extension to all integer targets, the result of Section 2 in finite and asymptotic forms, the strict terminal inequality, the scalar lemmas of the control quantization, and the soundness of the checkers for records, paths, and output profiles. The soundness proof shows that if the Boolean checks of a round accept its data, then every record of that round asserts a valid density statement. It proceeds by well-founded induction on the lexicographic order described above.

The generated material comprises the initial tangent bounds, the shared logarithm catalog, the support and profile checks, and the per-round guards, organized into eight round assemblies whose composed validity yields the final profile; the transfer and terminal modules then give Theorem 1.1. Each shared catalog enclosure is independently recomputed before it is shared. Table 1 collects the statistics.

B.4 Reproducing the verification

The full build uses the tagged repository together with the generated Lean sources, manifest and checksums attached to its GitHub release [9].

We pin the upstream repository of [12] at the commit `e53b8cf11d06`, and Mathlib correspondingly, and the shared upstream source modules were audited against those commits. The run reports, the resource and monitoring records, the source audit, and the reproduction commands are documented in the repository [9]; its driver script rebuilds the development against a matching upstream checkout and replays the final theorem and its axiom audit.

C Support conditions in the HorizonMath constructions

This appendix explains why the claimed upper bounds 3.7296^k and 3.6961^k , from version 1 of [15] and the repository snapshot [8], are not justified by their released verifications. Both constructions invoke a density theorem whose hypotheses include membership in the Ramsey region $\mathcal{R}$, and the supplied checks do not establish this hypothesis. For the 3.6961 construction, Neisler, Shin and Sukhatankar have observed independently that it fails their self-consistency inequality [12]; the

analysis here concerns the support conditions, and covers both constructions. The issue concerns these derivations, not the possibility of proving either numerical bound by another argument.

C.1 The density theorem

A refinement starts with an established off-diagonal upper bound

$$\log R(k, \ell) \leq kE(\ell/k) + o(k) \quad (1 \leq \ell \leq k).$$

From this bound it constructs admissible parameters (x, y) for a density theorem. Admissibility means that (x, y) belongs to the asymptotic Ramsey region $\mathcal{R}$: small inward perturbations yield $R(k, \ell) \leq x^{-k}y^{-\ell}$ for all sufficiently large $k + \ell$, in either order of the two targets.

The density theorem used in the released refinement is the case $\theta = 1$ of the candidate method [6, Theorem 13]. Its hypotheses include

$$(x, y) \in \mathcal{R}_*, \quad x < (1 - \mu)p^{1/(1-\mu)}. \quad (\text{C.1})$$

With these hypotheses, sufficiently large red-dense graphs of order at least $x^{-k/2}(\mu y)^{-\ell/2}$ have the required clique.

For a proposed exponent function F , the induction relates the density threshold to the derivative by $p \approx 1 - e^{-F'(t)}$. Writing m for the parameter μ , define, for $s > 0$ and $0 < m < 1$,

$$\chi(s, m) = (1 - m)(1 - e^{-s})^{1/(1-m)}.$$

The local parameter conditions are region membership, the slope condition $x < \chi(F'(t), m)$, and the size condition

$$F(t) + \frac{1}{2}\{\log x + t \log m + t \log y\} > 0. \quad (\text{C.2})$$

Together with the initial bound and the usual strictness and continuity conditions, these justify the refinement. Choosing x from the slope condition and checking (C.2) still leaves the region-membership hypothesis in (C.1) to be established.

C.2 The two support inequalities

Put $A = -\log x$ and $B = -\log y$. To establish the required region membership using the input profile E , the two target orders give respectively

$$A + tB \geq E(t), \quad B + tA \geq E(t) \quad (0 < t \leq 1). \quad (\text{C.3})$$

Indeed, the first inequality compares $Ak + B\ell$ with $kE(\ell/k)$ when $\ell \leq k$; the second compares the same expression with $\ell E(k/\ell)$ when $k \leq \ell$. Thus the two inequalities require

$$B \geq \max\{0, b_u(A), b_s(A)\}, \quad b_u(A) = \sup_t \frac{E(t) - A}{t}, \quad b_s(A) = \sup_t (E(t) - tA).$$

Both suprema are over $0 < t \leq 1$. Symmetry exchanges the two families; it does not allow either family to be discarded. Their common specialization at $t = 1$ is only the diagonal test $A + B \geq E(1)$.

Proposition C.1. *Replacing the maximum $\max\{b_u(A), b_s(A)\}$ by the minimum $\min\{b_u(A), b_s(A)\}$ does not establish (C.3).*

Proof. The two inequalities require $B \geq b_u(A)$ and $B \geq b_s(A)$ simultaneously. The minimum encodes their disjunction.

For a concrete example, take $E(t) = (1+t)\log(1+t) - t\log t$, $A = 1/100$, and $B = 2\log 2 - A$. Since $E'(t) = \log(1+1/t) > A$ on $(0, 1]$, the second supremum is $b_s(A) = E(1) - A = B$. The minimum test therefore accepts this point. At $t = 1/100$, however,

$$A + tB < \frac{1 + 2\log 2}{100} < \frac{\log 100}{100} < E(1/100),$$

so the first support inequality fails. □

The first solution printed in Appendix A.2 of [15] uses precisely this minimum:

$$\text{b_table}[i] = \max(0.0, \min(\text{bu}, \text{bs})).$$

The same minimum appears in the archived copy of `B_of_a` supplied with the repository snapshot [8]. This condition does not establish the two support inequalities.

C.3 The 3.6961 construction

The separate 3.6961 construction has a different parameter rule. The second solution printed in Appendix A.2 of [15] uses

$$F(t) = (1+t)\log(1+t) - t\log t + P(t)e^{-t},$$

$$P(t) = -0.25t + 0.033t^2 + 0.08t^3 - 0.0778t^5.$$

Here we write X, Y for the numerical parameters x, y , not for vertex sets. The construction chooses $Y = (1 - 0.0012) \min\{1, c_{\text{rel}}/X\}$, where the released decimal is

$$c_{\text{rel}} = 0.26292278641656774 \approx \frac{1}{4}e^{0.137/e}.$$

On the branch $c_{\text{rel}}/X < 1$, this imposes a condition on the product XY , or equivalently on $A + B$. Such a diagonal condition does not imply the full family (C.3).

Proposition C.2. *At the first interval of the released 3.6961 construction, set*

$$\begin{aligned} t_* &= 0.001, & m_* &= 0.001, \\ X_* &= \chi(F'(t_*), m_*), & Y_* &= (1 - 0.0012) \frac{c_{\text{rel}}}{X_*}. \end{aligned}$$

For $A = -\log X_*$, $B = -\log Y_*$, and $t_0 = 0.00229$, one has

$$A + t_0 B < 0.005342, \quad U(t_0) > 0.015642,$$

where U is the established input profile (5.1). In particular, this point violates the first support inequality for U .

Proof. Substitution gives

$$\begin{aligned} X_* &= 0.9977179\dots, & Y_* &= 0.2632079\dots, \\ A &= 0.0022846\dots, & B &= 1.3348109\dots \end{aligned}$$

The rational-interval calculation `horizon_witness.py` in the repository [9] proves the two displayed bounds with a gap exceeding 0.0103, for both the printed decimal c_{rel} and its indicated exponential value, so either interpretation gives the violation.

The construction’s own curve also satisfies

$$F(t_0) - U(t_0) = (0.003t_0^2 - 0.0778t_0^5)e^{-t_0} > 0.$$

Thus the same point fails the support test against F as well. The region hypothesis required to apply the density theorem is not established by either of these asserted inputs. $\square$

The two findings concern different released arguments. For 3.7296, the support validator checks an insufficient condition. For 3.6961, a specific parameter used in the construction violates the required profile-domination inequality by a fixed positive margin. Interval arithmetic applied to the remaining slope and size conditions cannot supply the missing region certificate. Consequently, the released verification does not justify either claimed Ramsey upper bound.

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